

AutoTerm-SST: Adaptive Terminology Memory with Zero Session-Time Setup for Streaming Speech Translation

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Abstract

Retrieval-augmented speech translation improves rare terminology, but systems such as RASST still assume that users provide a glossary before each session. That assumption is inconvenient for live talks, while merging every domain into one growing glossary eventually places too many irrelevant terms in the same retrieval and prompt competition.

We present AutoTerm-SST, an interactive system that removes session-time glossary upload and domain selection. It organizes a large offline terminology catalog into topic slices, matches recent streaming translation context to bilingual slice prototypes, and activates a compact multi-slice working set under fixed retrieval and prompt budgets. On ten real ACL and medicine talks, AutoTerm matches a known-domain reference in terminology accuracy (88.3% vs. 88.1%) and Stream-LAAL BLEU (45.09 vs. 45.19), while requiring no domain input. It improves substantially over no terminology retrieval and outperforms a one-million-entry merged glossary by 3.9 terminology-accuracy points and 2.30 BLEU. The live demo exposes active slices and retrieved terminology as the topic changes, providing zero-setup terminology-aware streaming translation.

1 Introduction

Simultaneous speech translation (SST) is increasingly used for technical talks, lectures, and live media. Modern speech-language models can translate long-form audio directly (Qwen Team, 2025; Ouyang et al., 2025), but rare methods, datasets, clinical concepts, and named entities remain difficult to infer from local acoustic context. Glossary retrieval addresses this problem by exposing relevant source–target term pairs to the language model before decoding.

RASST (Luo et al., 2026) shows that retrieval-augmented generation can improve out-of-distribution terminology in streaming translation. Its glossary, however, is fixed before a session. In practice, a listener may not know the talk domain in advance, and asking every user to find, upload, or select a glossary creates friction. A seemingly simple alternative is to merge all domains into a universal glossary. This works at moderate scale, but it is not a sustainable catalog policy: the retriever and prompt have fixed capacity, so irrelevant and conflicting mappings increasingly compete with useful evidence.

We introduce **AutoTerm-SST**, a system demo that moves glossary selection from user setup into the streaming loop. Operators register modular topic slices offline. During a session, a training-free router compares recent generated translation with bilingual slice prototypes and keeps several high-scoring slices inside a fixed term budget. Retrieval and language-model decoding remain unchanged; AutoTerm only controls which part of the catalog is allowed to compete for each prompt. Users neither upload a glossary nor select a domain.

This design provides a practical expansion path. New topic resources can be registered without retraining the translation model or forcing every session to search the entire catalog. The demo makes this behavior visible: users see the active working set, retrieved term pairs, and terminology-highlighted translation while audio is streaming.

Our contributions are:

- We present a zero-session-setup terminology-aware SST system built on RASST, with a live interface for inspecting active glossaries and retrieved evidence.
- We introduce a training-free, budgeted multi-slice memory that uses recent streaming translation context instead of user-provided domain metadata.

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- On a ten-talk ACL/medicine stream, we show the complete systems progression from no RAG, to known-domain and automatic working sets, to 100k and 1M universal glossaries; we also verify 32-session real-time serving.

2 Related Work

Streaming speech translation. SST systems translate partial speech while balancing quality and latency. Prefix-to-prefix policies make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), and SimulEval standardizes streaming evaluation (Ma et al., 2020). Recent speech-language models broaden the modeling choices for direct SST (Seamless Communication, 2023; Zhang et al., 2024; Qwen Team, 2025). AutoTerm-SST does not propose a new translation model; it contributes a terminology-management and demo layer around an existing streaming model.

Terminology-aware translation. Terminology control is commonly implemented with user-provided dictionaries, constrained decoding, or training-time exposure to constraints (Hokamp and Liu, 2017; Dinu et al., 2019). Terms and named entities remain a known weakness of speech translation (Gaido et al., 2021). Retrieval-augmented generation offers a non-parametric alternative (Lewis et al., 2020; Khandelwal et al., 2021), and RASST applies glossary retrieval before streaming LLM decoding (Luo et al., 2026). These approaches normally assume that the relevant glossary is known before translation. AutoTerm instead manages a budgeted cache of topic slices online.

Interactive translation systems. Serving engines provide continuous batching for language models (Kwon et al., 2023; Zheng et al., 2024), while interactive translation tools make model behavior inspectable (Dandekar et al., 2024). AutoTerm-SST combines streaming audio transport with a live terminology evidence view and automatic glossary selection.

3 System Overview

AutoTerm-SST wraps the RASST speech encoder, MaxSim term retriever, and Qwen3-Omni language model with an adaptive terminology loop (Figure 1). For chunk t , the speech encoder produces h_t and the retriever searches only the current working set. At most ten source–target term pairs are

placed in the language-model prompt. After decoding, the new translation is appended to a short context window and may change the working set for a later chunk.

The adaptive step is outside the translation model. It neither changes the speech representation nor grants the retriever a larger candidate budget. Instead, it decides which prebuilt indexes share that budget. This separation keeps the system pluggable: the same streaming protocol can serve a no-RAG agent, a fixed-glossary RASST agent, or AutoTerm, while the web interface receives structured evidence with every partial translation.

AutoTerm enforces three invariants: the live request contains no domain or glossary; catalog growth cannot increase per-chunk retrieval or prompt budgets; and a new topic is an offline data update, not model training. All policies therefore share the speech model, retriever, decoder, and streaming schedule, and differ only in which terminology indexes are eligible.

The live agent emits the translated text together with retrieved term pairs, their source slices, retrieval latency, and the current working-set state. These fields drive the evidence panel in Section 5 and make terminology behavior observable rather than hidden inside a prompt.

4 Adaptive Terminology Memory

AutoTerm separates the broad offline catalog from the compact glossary exposed to a live session. Each topic slice contains term–translation pairs, a prebuilt MaxSim index, and a bilingual prototype formed from a short topic description and representative terminology. Domain-dependent mappings are retained: the same source term may have different translations in different slices.

Offline catalog registration. An operator registers each slice through a manifest containing its label, term count, prototype, and prebuilt index. Slices are independently replaceable; topic-dependent mappings remain separate. At session time the router reads the compact manifest and prototypes, while indexes outside the working set do not participate in retrieval. Catalog capacity is thereby separated from the terminology exposed to one decoding step.

Semantic routing. At chunk t , the router concatenates the most recent w generated translation chunks into context c_t . For slice d with prototype

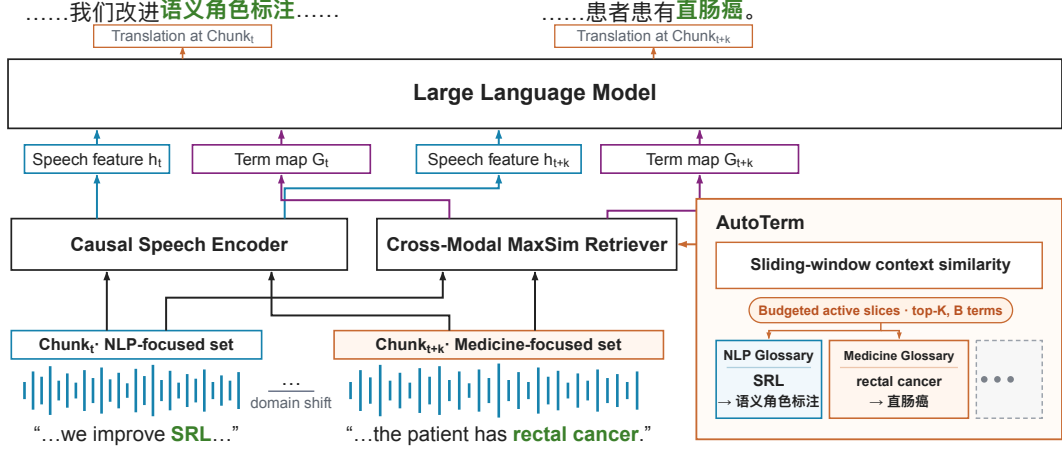


Figure 1: AutoTerm-SST as a stream moves from NLP to medicine. Recent translation context selects several glossary slices under a fixed term budget; RASST retrieves prompt evidence only from this working set. The ellipsis denotes additional registered topic slices.

examples P_d and multilingual sentence encoder E , it computes

$$\bar{p}_d = \text{norm} \left(\frac{1}{|P_d|} \sum_{p \in P_d} E(p) \right), \quad (1)$$

$$s(d | c_t) = \cos(E(c_t), \bar{p}_d).$$

We use BGE-M3 for E . Prototypes contain a short bilingual description and representative terms, so Chinese generated context and English topic metadata share one embedding space. The routing path is therefore a direct semantic similarity computation over accumulated translation context, not a classifier trained on the evaluation talks.

Why generated target context? The speech-language model translates audio directly, without a stable source transcript. Its partial target text is already available and becomes more topic-specific over time, enabling comparison with bilingual prototypes without a supervised domain model. Routing is causal: chunk t only changes later retrieval. A generic early translation can therefore delay topic evidence; accumulating four chunks and retaining several slices makes this signal less brittle.

The highest-scoring slices are admitted until either the maximum slice count K or total active-term budget B is reached. The current best slice is pinned when possible to reduce oscillation. Keeping multiple plausible slices is important: a temporary top-1 routing error need not remove the correct terminology from retrieval.

Working-set update. At each routing tick, AutoTerm keeps the current slice if it fits, then fills remaining slots in similarity order. Lower-ranked non-pinned slices stay outside the budget. This cache-like rule is not a hard domain switch: plausible topics can coexist during a transition, while stale slices disappear as their scores fall. Candidate indexes are preloaded and selections change only between decoding steps.

Fixed evidence budget. All active slices share 100 retrieval candidates and a common top-10 prompt cap. Increasing K therefore changes the composition of the search space, not the amount of evidence available to AutoTerm. In the evaluated configuration, the router considers ten 1k slices, uses the latest four translation chunks, and activates at most four slices ($B=4,000$). The known-domain reference searches the matching 1k slice; universal baselines search nested 100k and 1M indexes under exactly the same retrieval and prompt limits.

For a multi-slice query, the system divides the 100 candidate slots among the active indexes, combines their results, removes only exact duplicate source–target pairs, and globally reranks the remaining candidates before the top-10 prompt cap. Consequently, selecting four slices does not give AutoTerm four times the retrieval or prompt capacity. Its advantage can only come from placing more relevant mappings inside the same evidence budget.

The context encoder scores all lightweight prototypes together, and likely indexes are warmed

asynchronously. Each chunk retrieves from a snapshot of the working set installed before that chunk began; a routing update that finishes during generation only affects the next chunk. This keeps terminology evidence causal and prevents cold index loading from changing an in-flight prompt.

The million-entry baseline is a capacity stress condition, not a claim that large-index search is computationally impossible. AutoTerm addresses a different problem: preserving relevant prompt evidence as an offline catalog grows, without requiring any session-time input from the user.

5 Demo Interface

The demo begins like an ordinary streaming translator: the user selects a language pair and starts a microphone or file stream. In AutoTerm mode there is no glossary upload or domain selector. The system starts with a small working set and updates it as translation context accumulates.

Walkthrough. The page first shows live partials while context accumulates. The working-set badge then updates to the most similar slices, and retrieved pairs appear beside the partial that used them. After a topic change, old and new slices can briefly coexist until the sliding context reflects the new subject. The user changes no control throughout this progression.

Visible terminology evidence. The interface highlights retrieved terminology in the translation and lists the corresponding source–target mappings. A compact status row shows the active slices, term count, and retrieval latency. Thus a viewer can tell whether a mistranslation came from missing retrieval evidence, an irrelevant slice, or language-model decoding.

Extensibility. Operators register or replace topic slices offline; users see the benefit in the next session without model training or UI changes. Fixed-glossary presets and ad-hoc term overrides remain available for expert users, but they are not needed for the zero-setup workflow.

Registration validates term pairs, builds an independent index, and adds one manifest entry. Existing sessions retain their current snapshot, while new sessions immediately see the updated candidate set. Thus an organization can grow or revise its catalog without restarting the translation service or rewriting the client application.

Availability. The MIT-licensed implementation, reviewer-friendly launcher, and screencast are available in the [GitHub repository](https://luojiaxuan.github.io/autoterm-sst/). The hosted entry page is <https://luojiaxuan.github.io/autoterm-sst/>; live mode uses Qwen3-Omni with vLLM continuous batching and the RASST retriever.

6 Evaluation

Setup. We stream five complete ACL talks and five medicine talks through one alternating En→Zh session (approximately 4.7 hours). All terminology policies use 1.92s chunks, Qwen3-Omni, the same MaxSim retriever, 100 retrieval candidates, score threshold 0.78, and at most ten prompt references. We compare: *InfiniSST (no RAG)*, historical no-retrieval outputs on the same talks at the same chunk size (Ouyang et al., 2025); *Known-domain-1k*, which requires the talk domain; *AutoTerm-1k×4*, which requires no session input; and *Merged-100k/1M*, two nested universal glossaries.

The ten registered AutoTerm slices contain 1k entries each. The merged baselines combine the same target-bearing resources with distractors to exactly 100k and 1M mappings; 100k is a strict prefix of 1M. Exact duplicate pairs are removed, but conflicting domain-dependent translations are retained. Router settings were selected on four development windows before the complete run. Because the registered NLP and medicine slices include benchmark terminology, this is a controlled routing and retrieval-competition evaluation, not a held-out glossary-induction test.

Comparison provenance. The no-RAG row reuses the public *InfiniSST* hypotheses released with RASST for these exact ten talks and the matching 1.92s policy; it does not require another model run. We resegment those hypotheses against the same references and rescore all 2,047 term occurrences used by the RAG conditions. The other four rows come from one audited run per policy with identical audio order and chunk boundaries. Corpus BLEU is computed after pooling the resegmented sentences, rather than by averaging the ACL and medicine BLEU scores.

Metrics. **TERM_ACC** uses one fixed set of 2,047 time-aligned occurrences from the raw ACL tagged glossary and medicine hard/raw glossary. Exact-span source aliases with identical target variants are one spoken occurrence; nested terms



Figure 2: The live En→Zh interface after an automatic transition to medicine. Retrieved term pairs, source slices, and translation highlights expose the evidence used for each partial output.

Setting	Input	Active	TERM_ACC (%)			BLEU	MT-BLEU	Prec. (%)	Refs
			NLP	Med.	All				
InfiniSST (no RAG)	None	0	50.2	52.9	51.1	39.26	36.70	0.0	0.00
Known-domain-1k	Domain	1k	88.8	86.6	88.1	45.19	42.39	48.7	0.69
AutoTerm-1k×4	None	≤4k	89.3	86.3	88.3	45.09	42.28	34.8	0.94
Merged-100k	None	100k	89.4	84.5	87.8	43.72	40.93	6.8	4.96
Merged-1M	None	1M	87.1	78.9	84.4	42.79	39.95	3.4	8.70

Table 1: Ten-talk comparison under a fixed retrieval/prompt budget. BLEU and MT-BLEU use StreamLAAL-compatible mWER resegmentation. TERM_ACC uses the same 2,047 raw-term occurrences for every row (1,368 NLP; 679 medicine). Prec. is prompt precision. No RAG injects no references, so its prompt quantities are zero.

and different translations remain separate. Every condition, including no RAG, is rescored against this same denominator. **BLEU** follows the RASST/StreamLAAL pipeline: each talk hypothesis is resegmented to reference sentences with mWERSegmenter before corpus sacreBLEU. **MT-BLEU** applies the same pipeline after masking target translations from the corresponding corpus’s raw glossary. Prompt precision is the fraction of injected references that overlap a local raw-term occurrence; Refs is the mean number of references injected per chunk. The masking inventory is the full raw tagged set, not a curated subset used for interface highlights; we do not report a second filtered masked-BLEU variant.

From no RAG to an adaptive working set. Table 1 answers three systems questions. First, terminology retrieval matters: AutoTerm improves TERM_ACC from 51.1% without RAG to 88.3%, while BLEU rises by 5.83. Second, automatic routing removes setup at negligible quality cost. AutoTerm is within 0.24 TERM_ACC points and 0.11 BLEU/MT-BLEU of the known-domain reference, despite receiving no domain input. Third, a universal glossary is usable at 100k but degrades as it grows. Relative to Merged-100k, AutoTerm gains

1.37 BLEU and provides much cleaner evidence (34.8% vs. 6.8% prompt precision). At 1M, its advantage grows to 3.91 TERM_ACC points and 2.30 BLEU, while the merged baseline fills nearly the entire top-10 prompt with low-precision references.

Where does the scale loss occur? The domain columns in Table 1 prevent the aggregate result from hiding a useful qualification. Merged-100k is still competitive on NLP and loses only 1.8 points to AutoTerm on medicine. At 1M, however, the medicine gap expands to 7.4 points and an NLP gap also appears. Thus the result does not claim that every large index immediately fails. It shows that, under fixed retrieval and prompt capacity, the universal strategy becomes increasingly sensitive to the term distribution and ambiguity of each domain; a compact routed working set avoids that exposure.

Why does translation quality move? Prompt evidence exposes the mechanism behind the aggregate scores. AutoTerm injects fewer than one reference per chunk on average at 34.8% precision. Merged-100k injects 4.96 at 6.8%, and Merged-1M injects 8.70 at only 3.4%. Under the same threshold, the larger indexes fill a fixed prompt with more low-value mappings. MT-BLEU masks the eval-

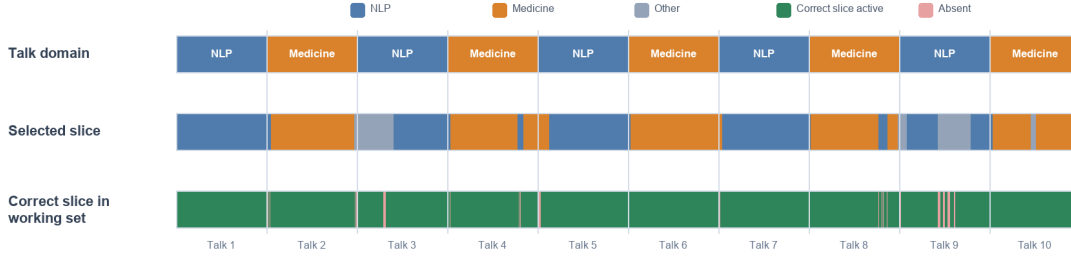


Figure 3: AutoTerm routing across ten alternating talks. The selected slice may briefly lag a topic change, but the correct glossary remains available inside the four-slice working set for 96.9% of decoder windows. Talk widths are normalized for readability.

Sessions	p50 (ms)	p95 (ms)	p95/stride
8	482	805	0.42
16	420	826	0.43
24	471	894	0.47
32	519	820	0.43

Table 2: End-to-end WebSocket serving on $2 \times A6000$ (TP=2). The service-time/1.92s-stride ratio stays below 0.5 through 32 sessions.

uated term translations, yet AutoTerm still leads Merged-1M by 2.33 points; irrelevant evidence therefore affects surrounding translation, not only tagged terms. Prompt precision exposes the mechanism, while TERM_ACC and MT-BLEU measure its consequences.

Routing behavior. Figure 3 illustrates why AutoTerm uses a multi-slice working set rather than a hard top-1 router. Transient top-slice errors occur around topic boundaries, yet another active slice usually preserves the correct terminology inventory. This limits the cost of delayed topic evidence in the generated translation context.

Observed failure modes. After one ACL-to-medicine boundary, the accumulated context retained the old topic for several chunks, so the medicine slice was temporarily absent before entering the four-slice working set. Lexical ambiguity creates a second error mode. In a medicine segment discussing scientific papers, Merged-1M supplied both the Chinese mapping for a scholarly article and an incompatible mapping for physical paper, whereas AutoTerm retained only the context-appropriate first sense. The example illustrates that routing reduces, rather than eliminates, ambiguous prompt evidence.

Concurrent serving. The hosted backend uses vLLM continuous batching. In an end-to-end real-time feed through the framework and WebSocket protocol, each client sends audio at the 1.92s input cadence and all 32 sessions complete without failure. The test includes scheduler admission, model generation, and WebSocket delivery; its generation p95 remains 0.82s at 32 sessions, or 0.43 of the input stride (Table 2). The nearly flat p95 from 8 to 32 clients is the expected continuous-batching behavior. We report this directly measured service-time ratio rather than inferring an end-to-end real-time factor from engine-only throughput.

7 Limitations and Conclusion

The main evaluation is En→Zh and uses ten registered topic slices. Those slices contain the benchmark terminology, so the results measure routing and retrieval competition rather than unseen-glossary induction. Routing also depends on generated target context and can lag after a topic boundary; the multi-slice budget only reduces this failure mode. The degradation threshold also depends on the language pair, retriever, ambiguity, and prompt budget; 100k is an intermediate point and 1M a stress condition, not a universal threshold. Fresh multilingual evaluation remains future work.

AutoTerm-SST turns terminology retrieval into a zero-session-setup system. Operators expand a modular offline catalog while users simply start a stream; recent context maintains a compact working set and RASST supplies prompt evidence. On mixed ACL and medicine talks, AutoTerm matches a known-domain reference, improves over no RAG, and avoids the quality and prompt-noise degradation of a 1M universal glossary. The demo makes this behavior visible and provides a practical path across many domains.

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